

What are AI solutions?

Stefan Schlobach

September 23, 2024

Recap

Lectures

Guest lectures (70% of the exam questions)

Still to come:

- Benno Kruit (Data processing and Engineering)
- Koen Hindriks (Social AI)
- Guszti Eiben (Computational Intelligence)
- Stefan Schlobach (Ethical AI)
- Ilaria Tiddi (Hybrid Intelligence)

Lectures

Methodology Lectures (30% of the exam questions)

Steps

- ~~1. Determining the problem~~
- ~~2. Finding Literature~~
- ~~3. Basics of computers~~
4. Methods in AI (approach/solution)
5. Empirical Evaluations
6. Mathematics for AI (Discrete)

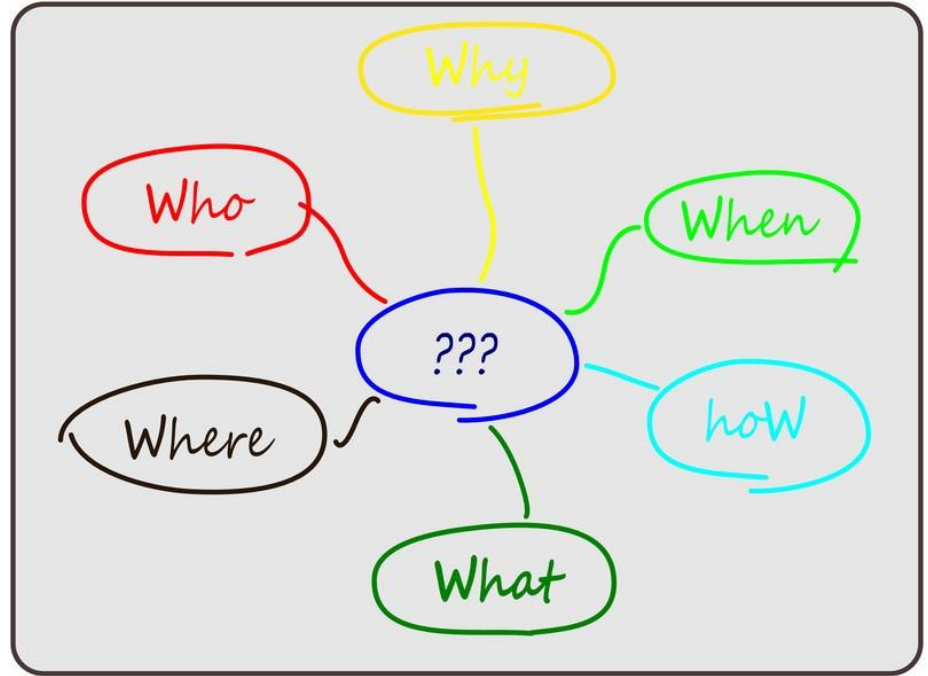
Where are we?

Where are we?

1. Defining the problem
2. Finding related work
3. **Building an AI system**
4. Evaluation of an AI system

1. Defining the Problem

- What is the **problem** we want to solve?
- **What** do we expect the system do do?



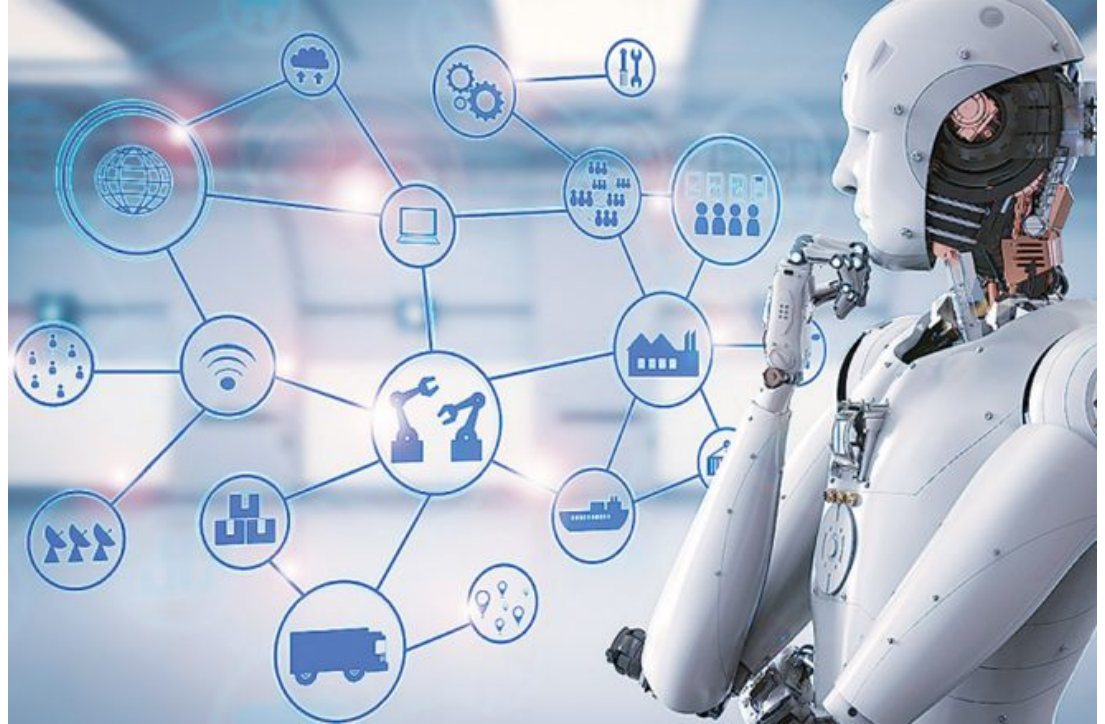
2. Finding Knowledge in Literature

- What **is known** in the literature about my problem?
- What have **others** tried to do?
- What are their **limitations**?



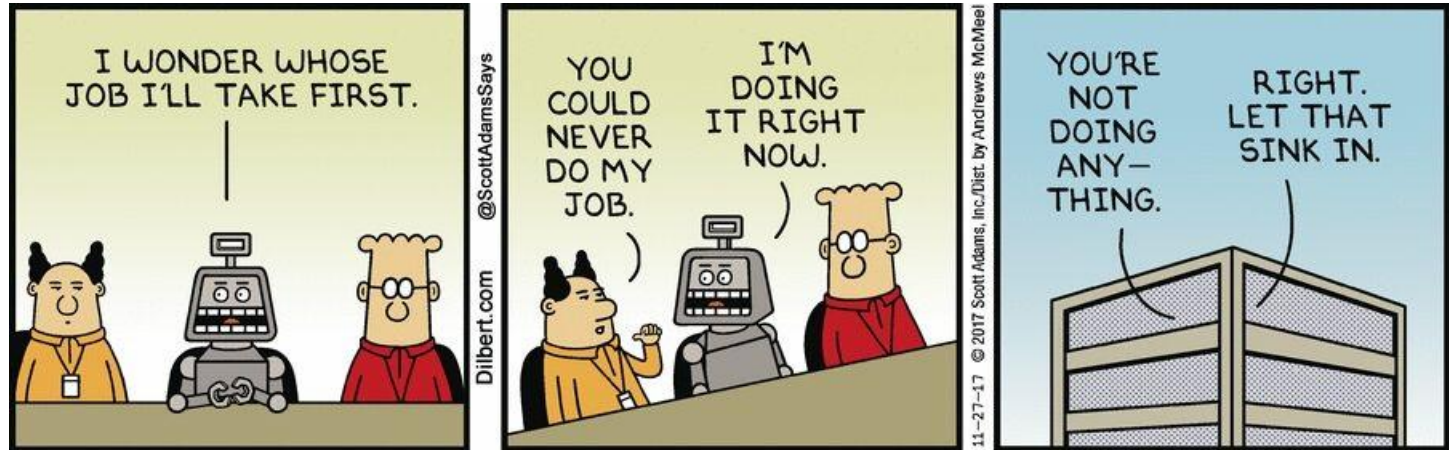
3. (Today) Building AI Systems

- What do you propose?
- Why? (motivation)
- How will your system work (substeps)?

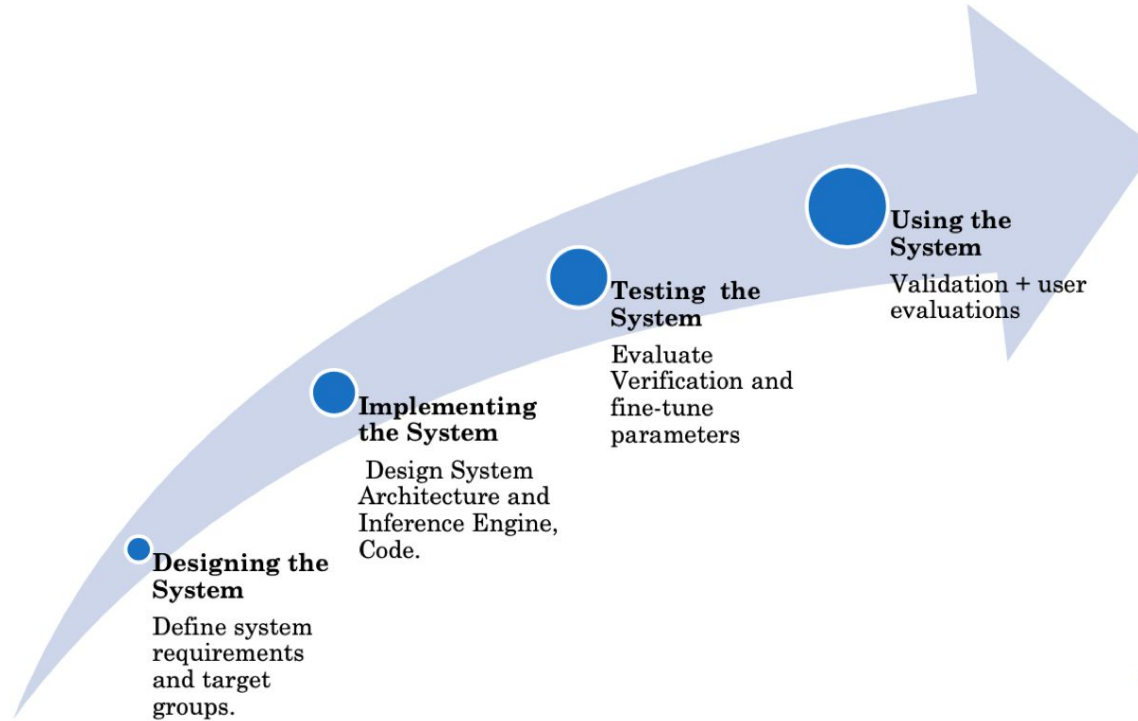


4. (to come) Evaluation of AI Systems

- How do I know I am done?
- Is my system acceptable?



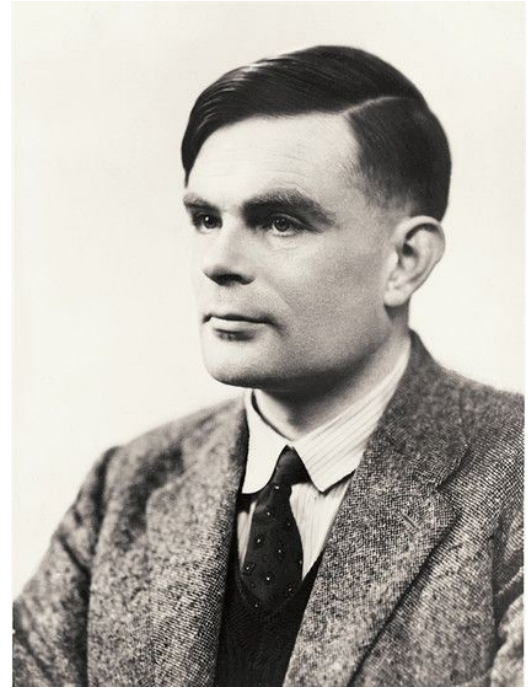
System Design



1. What is AI?

Quiz: who is this guy?

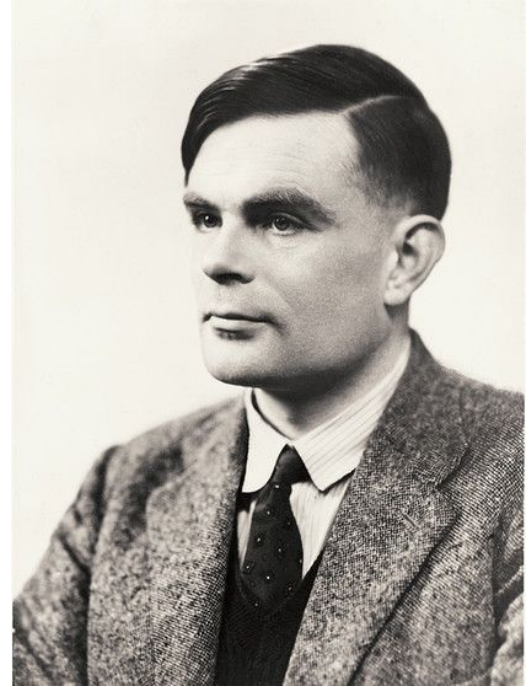
Why should you know him?



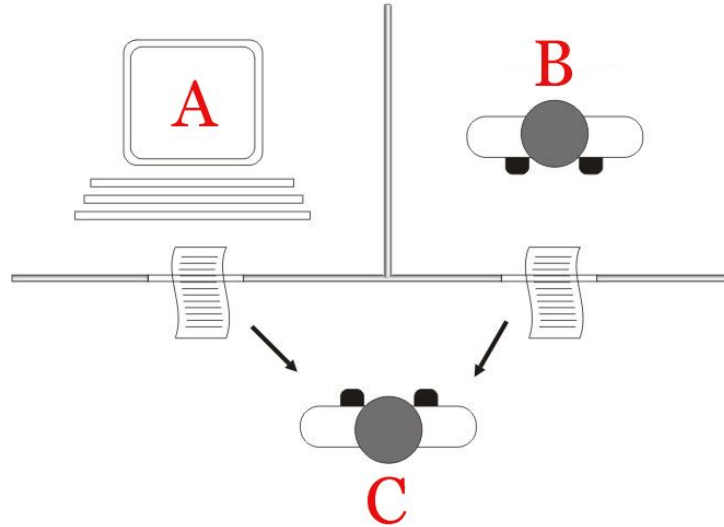
Quiz: who is this guy?

*“AI is the science and engineering of **making intelligent machines**, especially intelligent computer programs.”*

Alan Turing

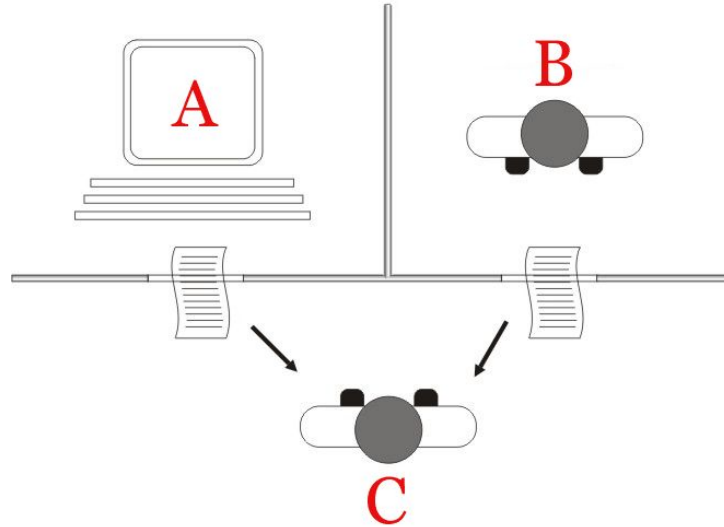


Human Intelligence



Human Intelligence

Only defines
how to
recognise
intelligence

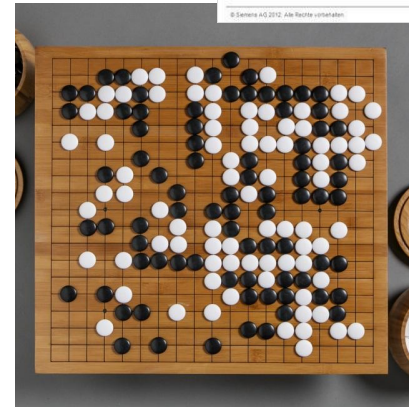
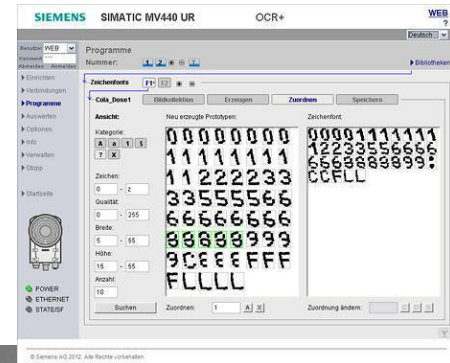


Only defines
if the machine
behaves as a
human

Complex Algorithmics

Current AI **do not fit the definition** (only operated in a narrow space):

- Optical Character Recognition
- Speech recognition
- Automated Assistants (Siri, Alexa)
- Autonomous driving
- Chess
- Go
- Question Answering



Complex Algorithmics

Horizon effect/problem

- Search space (the number of possible states/positions) is immense
- Feasible search on a small portion of states
- Possibility to make a detrimental move without knowing the effect (i.e. beyond its "horizon").



Complex Algorithmics

Missing **rationality**!

An “Intelligent Agent” is a device that **perceives** its environment and **takes actions** that maximise its chance of successfully **achieving its goals**.



Extending the Definition of Intelligence



Ability to **plan, reason and learn, sense** and build some kind of **perception** of knowledge and **communicate** in natural language.

2. Types of AI

Types of AI

Categorisation of AI methods:

- capacity to mimic human characteristics
- the technology they use
- the real-world application
- the theory of mind

Types of AI

1. Artificial narrow intelligence (ANI), which has a narrow range of abilities;
2. Artificial general intelligence (AGI), which is on par with human capabilities; or
3. Artificial superintelligence (ASI), which is more capable than a human.

Narrow/Weak AI

- Limited abilities
- Goal-oriented (perform singular tasks)
- search, facial recognition, speech recognition/voice assistants, self-driving cars
- Narrow : small set of constraints and limitations to simulate human behaviour

Narrow/Weak AI

- Speech and language recognition (Siri virtual assistant)
- Vision recognition of self-driving cars
- Recommendation engines for products

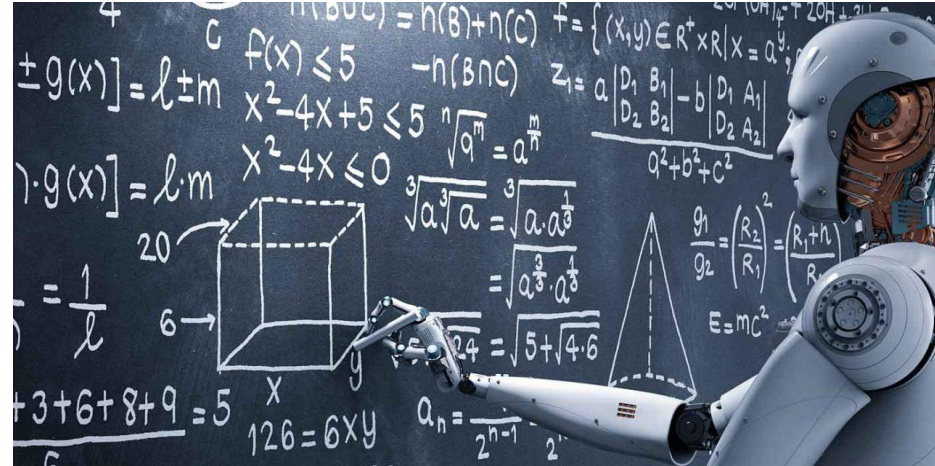
These systems only learn to complete specific tasks!

General/Strong AI

- mimics human intelligence and behaviours
- ability to learn to solve any problem
- Machines think, understand, and act indistinguishably from a human

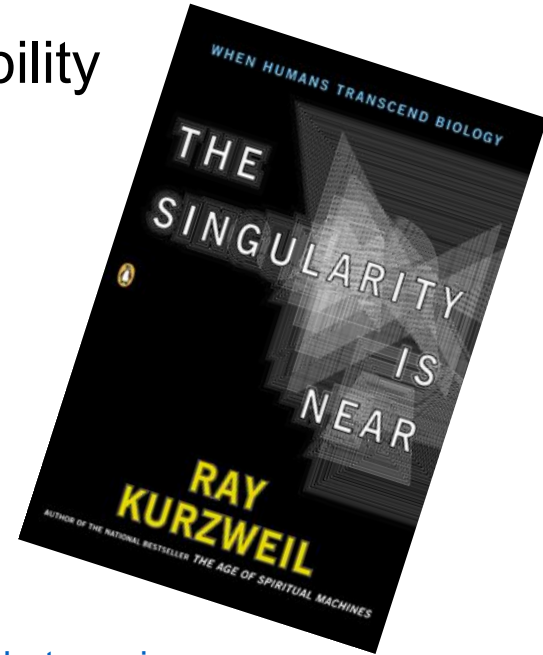
Yet to be achieved

lack of comprehensive knowledge on the functionality of the human brain



Super AI

- machines surpass the capacity of human ability
- Currently only in dystopian fiction
- Pure speculation



Further reading

<http://www.bbc.com/news/technology-30290540>

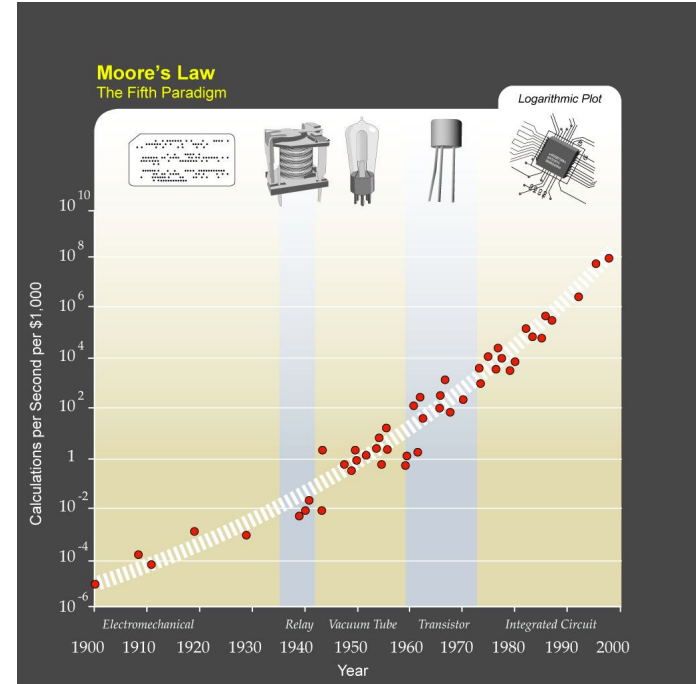
<http://www.theverge.com/2016/6/2/11837566/elon-musk-one-ai-company-that-worries-me>

<http://www.theverge.com/2015/12/11/9910742/elon-musk-openai-sam-altman-y-combinator-machine-learning>

Where are we going?

Rate of progress

- Stone tools (2 600 000 ya)
- Hafting (500 000 ya)
- Agriculture (~10 000 ya)
- Writing (6 000 ya)
- Steam engine (200 ya)
- Flight (100 ya)
- Space flight (50 ya)



Interpretable AI

- Can we build a Strong AI in a way that we understand?
- Interpretable weak AI:
- Do we understand Go any better?
- Do we understand vision any better
- Do aeroplanes help us understand birds?

3. AI Problems & typical related solutions

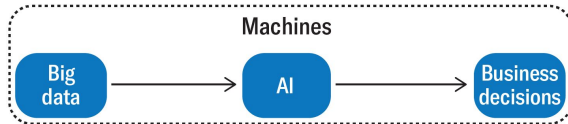
Important Note

- The following overview is very simplified.
- Usually an AI system involves a combinations of approaches

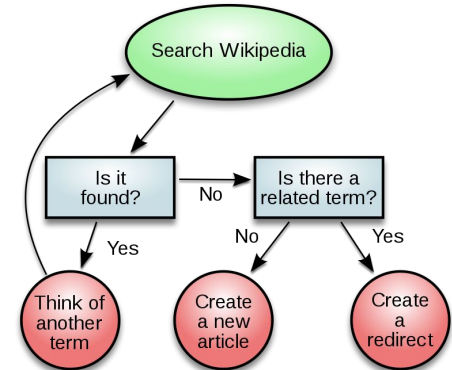
Problem solving : decision-making

- AI into the workflow as a primary processor of data
- Avoid human cognitive bias
- Dealing with large data

A Decision-Making Model That Utilizes AI



Adding an article to Wikipedia



Problem solving : decision-making

- systematic search through a range of possible actions in order to reach some predefined goal or solution
- specific-purpose methods:
 - tailor-made for a particular problem
 - exploits very specific features of the situation
- general-purpose methods
 - applicable to a wide variety of problems



Problem solving : Heuristics

- When finding the guaranteed best possible solution using exact methods is too complex
- a practical method for finding a sub-optimal solution
- s.o. = not optimal, but sufficient for the immediate goal
- Example : vehicle routing

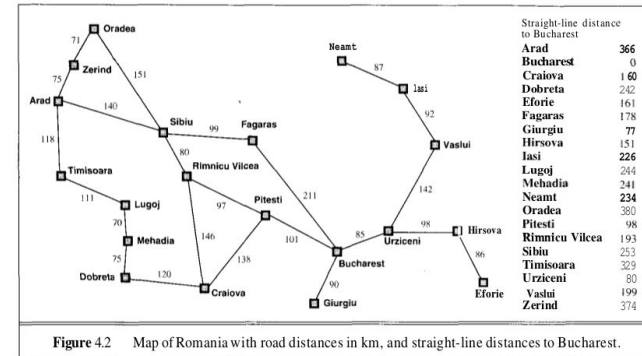
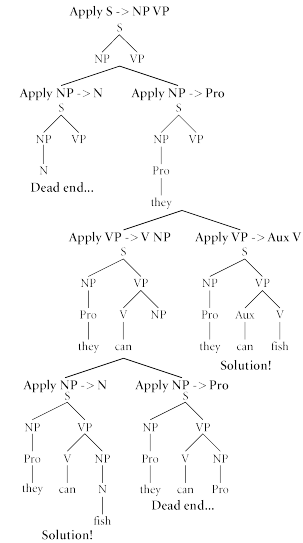
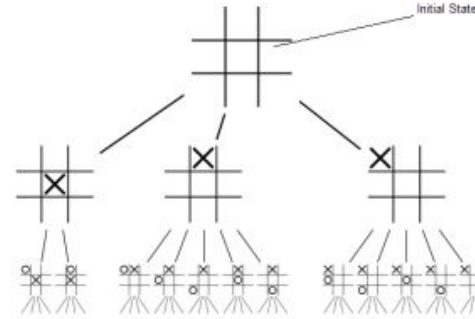


Figure 4.2 Map of Romania with road distances in km, and straight-line distances to Bucharest.

Learning

Machine learning (ML) is the [scientific study](#) of [algorithms](#) and [statistical models](#) that [computer systems](#) use to perform a specific task without using explicit instructions, relying on patterns and [inference](#) instead.

1. Supervised learning (Classification, Regression)
2. Unsupervised (Clustering)
3. Reinforced

Reasoning

Reason is the capacity of [consciously](#) making sense of things, establishing and verifying [facts](#), applying [logic](#), and adapting or justifying practices, [institutions](#), and [beliefs](#) based on new or existing [information](#)

- Deductive: produce new explicit statements from implicit ones.
- Inductive: produce explicit statements for unobserved objects.
- Abductive: produce statements that explains known statements.
- Case-based: solve problems based on solutions to similar problems

Reasoning : Knowledge Representation

Making sense of or explaining the world

- Interpreting perceptions
- Organising knowledge
- Conceptualising phenomena



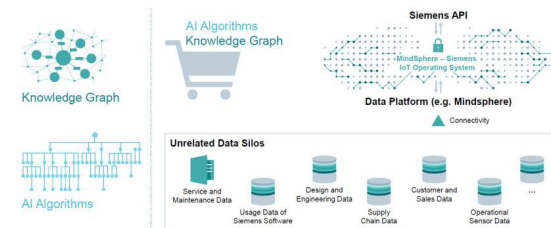
Useful for:

- Question Answering
- Decision support
- Data Integration
- Disambiguation



Knowledge as a Service – Consuming knowledge and analytics should be as easy as shopping

SIEMENS
Ingenuity for life.



Unrestricted © Siemens AG 2004

Reasoning : Logics

- Propositions
- Predicates
- Objects and instances / Graphs (Description Logics)
- Modal logics (belief, epistemic)
- Temporal
- Non-monotonic / Default
- Fuzzy logics
- Probabilistic

Perception

- The process whereby sensory stimulation is translated into organized experience
- environment scanned by various sensory organs
- scene is decomposed
 - separate objects
 - spatial relationships
- Example applications : autonomous vehicles , trash-picking robots

Computer Vision

How to gain high-level understanding from [digital images](#) or [videos](#)

- Object detection
- Object recognition
- Object Identification
- Motion analysis
- Event detection
- Emotion detection

Robotics



Automated Planning and Scheduling

Realisation of strategies or action sequence

Given a description of the possible initial states of the world, a description of the desired goals, and a description of a set of possible actions, the planning problem is to synthesise a plan that is guaranteed (when applied to any of the initial states) to generate a state which contains the desired goals (such a state is called a goal state).

Social Intelligence

Human-Robot Interaction

AI, psychology, human-computer interaction

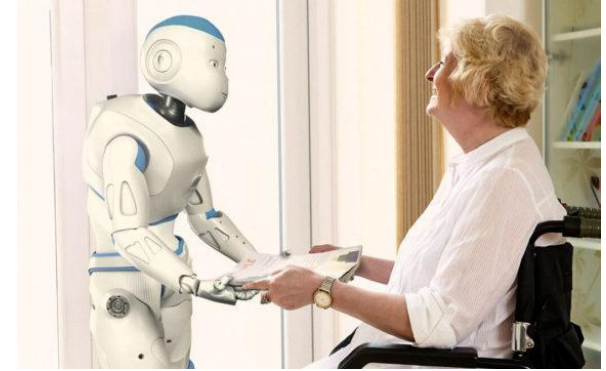
Machine ethics, moral agency

Methods for perceiving

Methods for

Cognitive models (theory of mind)

Healthcare, search&rescue,space exploration



Natural Language Processing

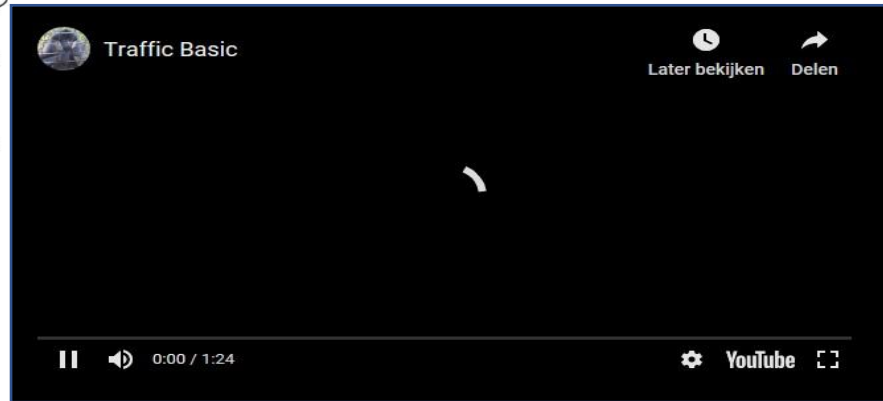
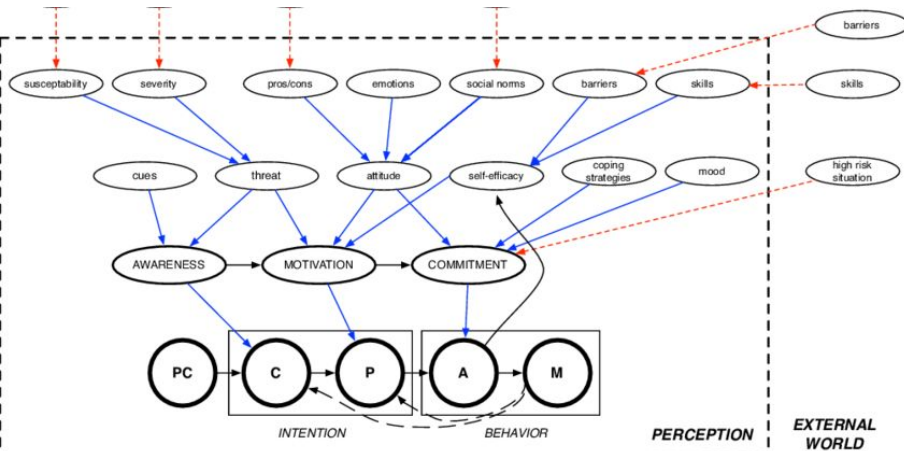
Language : a system of signs having meaning by convention
(e.g. spoken words, traffic signs)

Natural language processing

- Understanding the structure of natural language (Syntax)
- Understanding the meaning of natural language (Semantics)
- Understanding the usage of natural language (Pragmatics)
- Language generation
- Machine Translation

Engineering : Modelling and simulation

the use of conceptual models (e.g., physical, mathematical, or logical representation of a system, entity, phenomenon, or process) as a basis for simulations to develop data utilized for managerial or technical decision making [[source](#)]



Engineering : Model-based Diagnosis

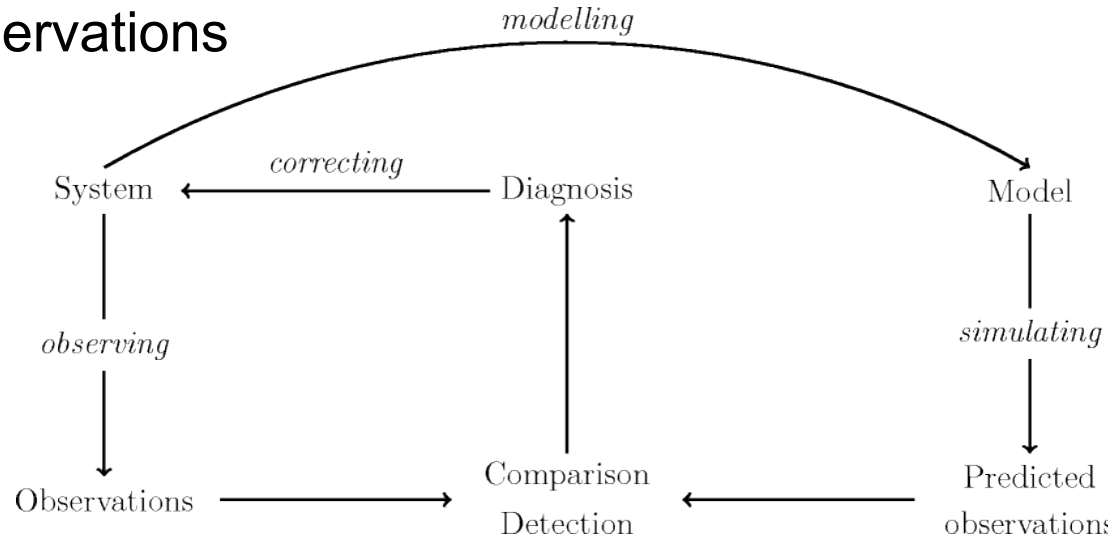
Goal : detect faulty behaviour

Formal model describes the behaviour of a system

The diagnosis system simulates the system using the model, and compares the real observations and the predicted ones.

Related to
Abductive reasoning

Applied in Healthcare .



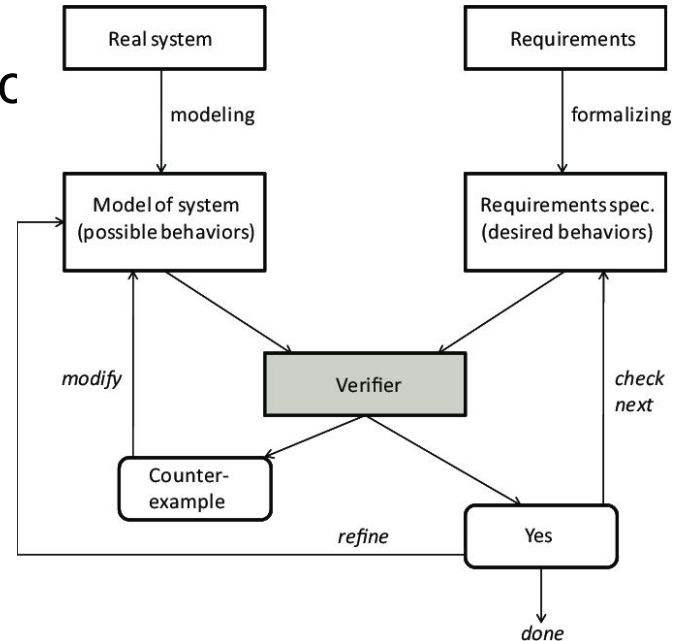
Formal Verification by Model Checking

checking whether a finite-state model of a system meets a given specific (= correctness)

Systems are modeled by finite state machines

Properties are written in mathematical forms

Verification = an exhaustive search of the state space of the design



Ethical AI

A branch of the ethics of technology specific to artificially intelligent systems.[\[1\]](#)

- moral behavior of humans as they design, make, use and treat AI systems
 - Algorithmic bias
 - Transparency, accountability, open source
- behavior of machines
 - robot ethics, machine ethics
 - Robot rights

Hybrid Intelligence



How to

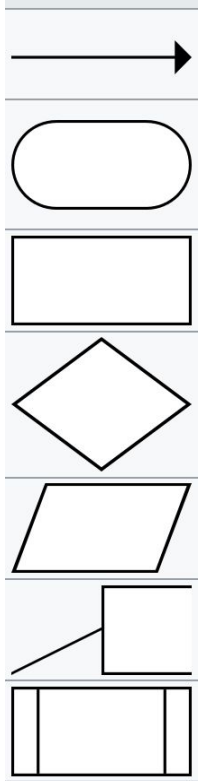
Designing an Approach for your task

- Describe the (sub)steps of your approach
- Expected:
 - a **flowchart** of your system
 - A step-by-step **description** of the chosen AI method to solve your problem
 - a **motivation** for your choice
 - A link to the literature and **why this is this novel** (contribution)

Flowcharts

- A type of diagram that represents a process
 - a diagrammatic representation of an algorithm
 - allows to graphically describe the (sub)steps of your approach
- Diagram:
 - steps as different type of boxes
 - ordering of steps by arrows
 - inputs&outputs
 - Need to be a sketch of the bigger picture!
- Tools to use? Diagram.net, pen&paper...

Flowcharts



Flowline

Terminal (ending of program or subprogram)

Process

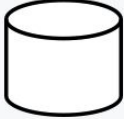
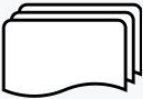
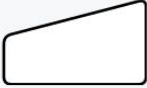
Decision

Input/Output

Comment

Predefined process (process is defined elsewhere)

Flowcharts

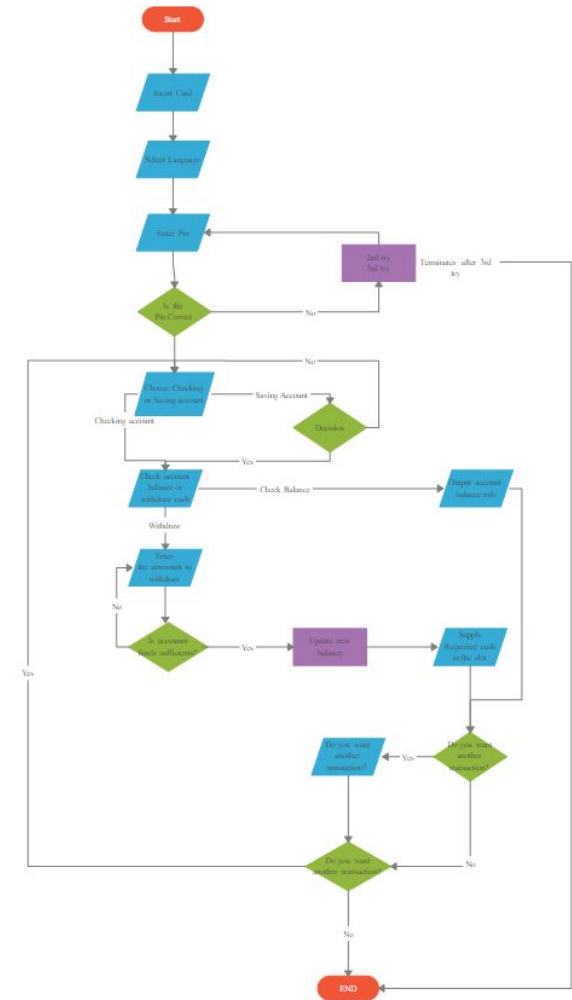
Shape	Name
	Data File or Database
 	Document
	Manual operation
	Manual input
	Preparation or Initialization

Flowchart example

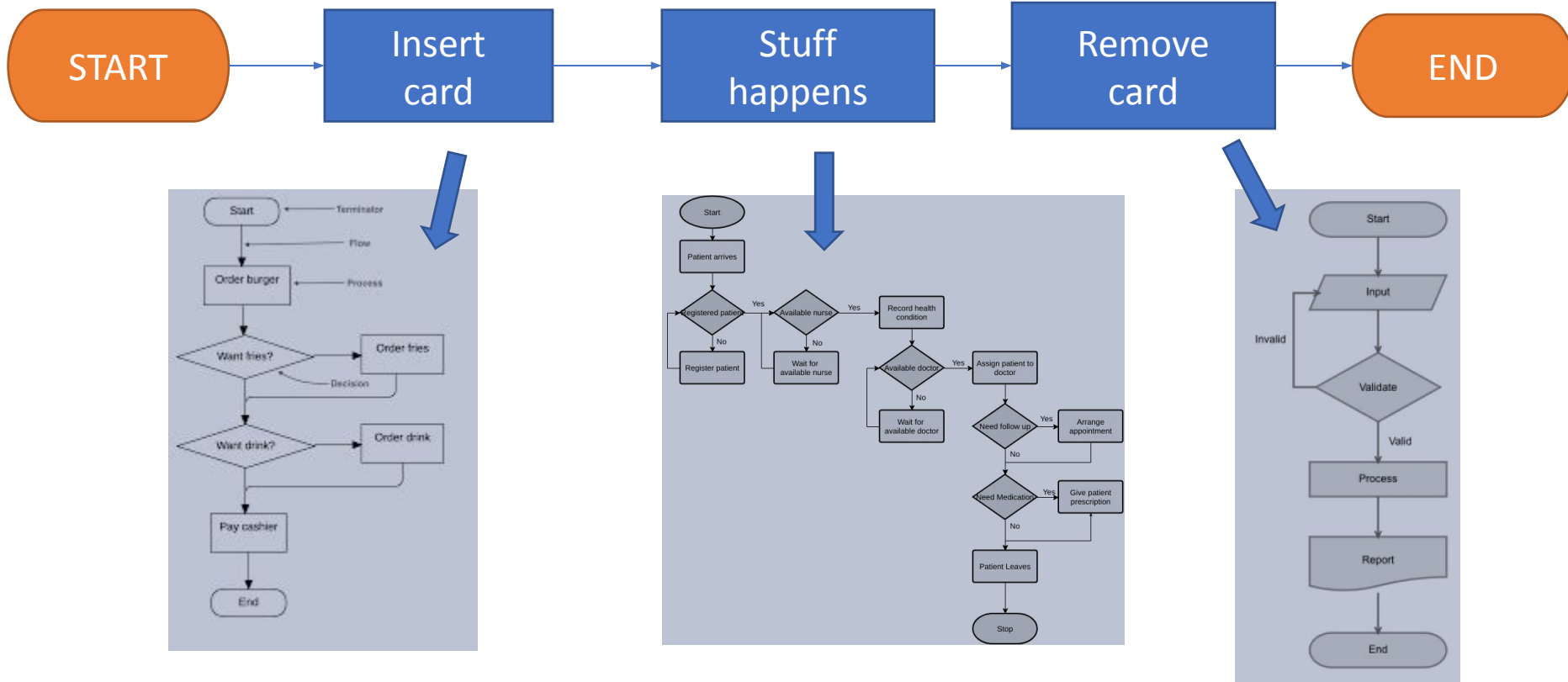


Different levels of abstraction

- Check account
- Is the PIN correct ?
- Enter amount to withdraw...



Top-down refinement



This week's tutoring session

- Literature Review
 - Get feedback on Literature Review
 - Integrate the comments about the literature review
- Approach/Computational Solution
 - Half-way Presentations (with tutors)
 - Work on Approach/computational solution (incl. workflows)
 - Submit the approach assignment

Coming Lectures

This Week

- (Friday) Guest Lecture : Data Engineering by Benno Kruit

Next Week

- (Monday) Methodology: Empirical Evaluations by Sara Lusini
- (Wednesday) Guest Lecture : Social AI by Koen Hindriks